

MTH 202 Chapter 1: Three-Dimensional Space & Vectors

1.4 Worksheet

1. Find the cross product $\mathbf{a} \times \mathbf{b}$ and verify that it is orthogonal to both $\mathbf{a}$ and $\mathbf{b}$.

a) $\mathbf{a} = \langle 6, 0, -2 \rangle$ & $\mathbf{b} = \langle 0, 8, 0 \rangle$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 6 & 0 & -2 \\ 0 & 8 & 0 \end{vmatrix} = \langle 0 - (-16), -(0 - 0), 48 - 0 \rangle = \langle 16, 0, 48 \rangle$$

Note that $\mathbf{a} \cdot \langle 16, 0, 48 \rangle = (6)(16) + (0)(0) + (-2)(48) = 0$ and

$$\mathbf{b} \cdot \langle 16, 0, 48 \rangle = (0)(16) + (8)(0) + (0)(48) = 0.$$

Since the dot products are equal to 0, we can see that the cross product is orthogonal to both $\mathbf{a}$ and $\mathbf{b}$.

b) $\mathbf{a} = \mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$ & $\mathbf{b} = -\mathbf{i} + 5\mathbf{k}$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 3 & -2 \\ -1 & 0 & 5 \end{vmatrix} = \langle 15 - 0, -(5 - 2), 0 - (-3) \rangle = \langle 15, -3, 3 \rangle$$

Note that $\mathbf{a} \cdot \langle 15, -3, 3 \rangle = (1)(15) + (3)(-3) + (-2)(3) = 0$ and

$$\mathbf{b} \cdot \langle 15, -3, 3 \rangle = (-1)(15) + (0)(-3) + (5)(3) = 0.$$

Since the dot products are equal to 0, we can see that the cross product is orthogonal to both $\mathbf{a}$ and $\mathbf{b}$.

c) $\mathbf{a} = \left\langle t, 1, \frac{1}{t} \right\rangle$ & $\mathbf{b} = \langle t^2, t^2, 1 \rangle$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ t & 1 & \frac{1}{t} \\ t^2 & t^2 & 1 \end{vmatrix} = \langle 1 - t, -(t - t), t^3 - t^2 \rangle = \langle 1 - t, 0, t^3 - t^2 \rangle$$

Note that $\mathbf{a} \cdot \langle 1 - t, 0, t^3 - t^2 \rangle = (t)(1 - t) + (1)(0) + \left(\frac{1}{t}\right)(t^3 - t^2) = (t - t^2) + (t^2 - t) = 0$ and

$$\mathbf{b} \cdot \langle 1 - t, 0, t^3 - t^2 \rangle = (t^2)(1 - t) + (t^2)(0) + (1)(t^3 - t^2) = (t^2 - t^3) + (t^3 - t^2) = 0.$$

Since the dot products are equal to 0, we can see that the cross product is orthogonal to both $\mathbf{a}$ and $\mathbf{b}$.

2. State whether each expression is defined or not. If not, explain why. If so, state whether it is a vector or a scalar.

a) $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$

This expression is defined. $\mathbf{b} \times \mathbf{c}$ results in a vector and we can then calculate a dot product between $\mathbf{a}$ and $\mathbf{b} \times \mathbf{c}$. This expression results in a scalar.

b) $\mathbf{a} \times (\mathbf{b} \cdot \mathbf{c})$

This expression is meaningless. $(\mathbf{b} \cdot \mathbf{c})$ results in a scalar and we can't compute a cross product between the vector $\mathbf{a}$ and a scalar.

c) $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$

This expression is defined. $\mathbf{b} \times \mathbf{c}$ results in a vector and we can then calculate a cross product between $\mathbf{a}$ and $\mathbf{b} \times \mathbf{c}$. This expression results in a vector.

d) $\mathbf{a} \cdot (\mathbf{b} \cdot \mathbf{c})$

This expression is meaningless. $(\mathbf{b} \cdot \mathbf{c})$ results in a scalar and we can't compute a dot product between the vector $\mathbf{a}$ and a scalar.

e) $(\mathbf{a} \cdot \mathbf{b}) \times (\mathbf{c} \cdot \mathbf{d})$

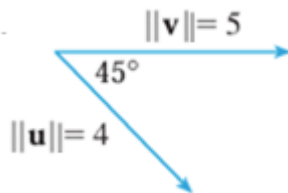
This expression is meaningless. $(\mathbf{a} \cdot \mathbf{b})$ and $(\mathbf{c} \cdot \mathbf{d})$ result in scalars and we can't compute a cross product between two scalars.

f) $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d})$

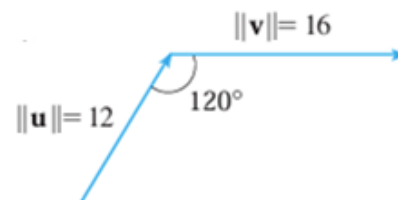
This expression is defined. $(\mathbf{a} \times \mathbf{b})$ and $(\mathbf{c} \times \mathbf{d})$ result in vectors and we can then compute a dot product between these two vectors. This expression results in a scalar.

3. Find $\|\mathbf{u} \times \mathbf{v}\|$ and determine whether $\mathbf{u} \times \mathbf{v}$ is directed into the page or out of the page.

a)



b)



We know that $\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin(\theta)$ and so,

a) $\|\mathbf{u} \times \mathbf{v}\| = (4)(5) \sin(45^\circ) = 20 \left(\frac{\sqrt{2}}{2} \right) = 10\sqrt{2}$. $\mathbf{u} \times \mathbf{v}$ comes *out* of the page.

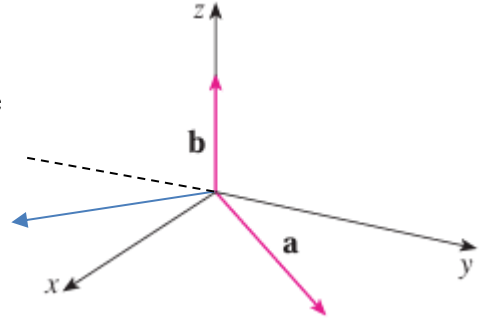
b) $\|\mathbf{u} \times \mathbf{v}\| = (12)(16) \sin(120^\circ) = 192 \left(\frac{\sqrt{3}}{2} \right) = 96\sqrt{3}$. $\mathbf{u} \times \mathbf{v}$ goes *into* the page.

4. The figure shows a vector $\mathbf{a}$ in the xy -plane and a vector $\mathbf{b}$ in the direction of $\mathbf{k}$. Their lengths are $\|\mathbf{a}\| = 3$ and $\|\mathbf{b}\| = 2$.

- a) Find $\|\mathbf{a} \times \mathbf{b}\|$.

The angle between $\mathbf{a}$ and $\mathbf{b}$ is 90° since $\mathbf{b}$ is perpendicular to the xy -plane. Therefore,

$$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin(\theta) = (3)(2) \sin(90^\circ) = 6(1) = 6$$



- b) Use the right-hand rule to decide whether the components of $\mathbf{a} \times \mathbf{b}$ are positive, negative or 0.

$\mathbf{a} \times \mathbf{b}$ is the blue vector shown in the picture above. We can see that it has a positive x -component, negative y -component, and a z -component of 0.

5. Consider the vectors $\mathbf{a} = \langle 1, 2, 3 \rangle$ & $\mathbf{b} = \langle -1, 1, 2 \rangle$

- a) Find the area of the parallelogram determined by these vectors.

We know that the area of the parallelogram determined by the vectors $\mathbf{a}$ and $\mathbf{b}$ is given by $\|\mathbf{a} \times \mathbf{b}\|$.

$$\text{We calculate that } \mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 3 \\ -1 & 1 & 2 \end{vmatrix} = \langle 4 - 3, -(2 - (-3)), 1 - (-2) \rangle = \langle 1, -5, 3 \rangle.$$

$$\text{Therefore, } \|\mathbf{a} \times \mathbf{b}\| = \sqrt{(1)^2 + (-5)^2 + (3)^2} = \sqrt{35}$$

- b) Find the area of the triangle determined by these vectors.

We know that the area of the parallelogram determined by the vectors $\mathbf{a}$ and $\mathbf{b}$ is given by $\frac{1}{2} \|\mathbf{a} \times \mathbf{b}\|$.

$$\frac{1}{2} \|\mathbf{a} \times \mathbf{b}\| = \frac{1}{2} \sqrt{35}$$

- c) Determine if $\mathbf{a}$ and $\mathbf{b}$ are coplanar with the vector $\mathbf{c} = \langle 2, 1, 4 \rangle$.

We need to compute the triple scalar product $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$. We already know that $\mathbf{a} \times \mathbf{b} = \langle 1, -5, 3 \rangle$ and so $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \langle 1, -5, 3 \rangle \cdot \langle 2, 1, 4 \rangle = 2 + (-5) + 12 = 9$.

Since $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = 9$ this implies that the volume of the parallelepiped determined by $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ has a non-zero volume and this implies that $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are not coplanar.

6. Consider the vectors $\mathbf{a} = \mathbf{i} + \mathbf{j}$ & $\mathbf{b} = \mathbf{j} + \mathbf{k}$

a) Find the area of the parallelogram determined by these vectors.

We know that the area of the parallelogram determined by the vectors $\mathbf{a}$ and $\mathbf{b}$ is given by $\|\mathbf{a} \times \mathbf{b}\|$.

$$\text{We calculate that } \mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 0 \\ 0 & 1 & 1 \end{vmatrix} = \langle 2-0, -(1-0), 1-0 \rangle = \langle 2, -1, 1 \rangle.$$

$$\text{Therefore, } \|\mathbf{a} \times \mathbf{b}\| = \sqrt{(2)^2 + (-1)^2 + (1)^2} = \sqrt{6}$$

b) Find the area of the triangle determined by these vectors.

We know that the area of the parallelogram determined by the vectors $\mathbf{a}$ and $\mathbf{b}$ is given by $\frac{1}{2}\|\mathbf{a} \times \mathbf{b}\|$.

$$\frac{1}{2}\|\mathbf{a} \times \mathbf{b}\| = \frac{1}{2}\sqrt{6}$$

c) Determine if $\mathbf{a}$ and $\mathbf{b}$ are coplanar with the vector $\mathbf{c} = \mathbf{i} + \mathbf{j} - \mathbf{k}$.

We need to compute the triple scalar product $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$. We already know that $\mathbf{a} \times \mathbf{b} = \langle 2, -1, 1 \rangle$ and so

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \langle 2, -1, 1 \rangle \cdot \langle 1, 1, -1 \rangle = 2 + (-1) + (-1) = 0.$$

Since $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = 0$ this implies that the volume of the parallelepiped determined by $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ has a zero volume and this implies that $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are coplanar.

7. If $\mathbf{a} \cdot \mathbf{b} = \sqrt{3}$ and $\mathbf{a} \times \mathbf{b} = \langle 1, 2, 2 \rangle$, find the angle between $\mathbf{a}$ and $\mathbf{b}$.

We know that $\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\|\|\mathbf{b}\|\cos(\theta)$ and $\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\|\|\mathbf{b}\|\sin(\theta)$ where θ is the angle between the vectors.

By combining these equations, we can say

$$\frac{\|\mathbf{a} \times \mathbf{b}\|}{\mathbf{a} \cdot \mathbf{b}} = \frac{\|\mathbf{a}\|\|\mathbf{b}\|\sin(\theta)}{\|\mathbf{a}\|\|\mathbf{b}\|\cos(\theta)} \rightarrow \frac{\|\mathbf{a} \times \mathbf{b}\|}{\mathbf{a} \cdot \mathbf{b}} = \tan(\theta) \rightarrow \frac{\sqrt{9}}{\sqrt{3}} = \tan(\theta) \rightarrow \theta = \tan^{-1}(\sqrt{3}) \rightarrow \theta = \frac{\pi}{3}$$

8. Suppose vector $\mathbf{v}$ is a vector in $\mathbb{R}^3$.

a) Find all vectors $\mathbf{v}$ such that $\langle 1, 2, 1 \rangle \times \mathbf{v} = \langle 3, 1, -5 \rangle$.

$$\text{Suppose that } \mathbf{v} = \langle a, b, c \rangle, \text{ then } \langle 1, 2, 1 \rangle \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 1 \\ a & b & c \end{vmatrix} = \langle 2c - b, -(c - a), b - 2a \rangle.$$

$$\text{In order for } \langle 2c - b, -(c - a), b - 2a \rangle = \langle 3, 1, -5 \rangle \text{ we must have } \begin{cases} 2c - b = 3 \\ -(c - a) = 1 \\ b - 2a = -5 \end{cases}$$

$$\text{By adding the first and third equations we get } \begin{array}{r} 2c - b = 3 \\ + b - 2a = -5 \\ \hline 2c - 2a = -2 \end{array} \rightarrow c - a = -1.$$

$$\text{By adding this new equation to the second equation we get } \begin{array}{r} c - a = -1 \\ + -(c - a) = 1 \\ \hline 0 = 0 \end{array}. \text{ This implies that we have}$$

a dependent system which means that there are multiple solutions and the values of a , b , and c all depend on one another.

If we return to the third equation we see that $b = 2a - 5$ and by the second equation we have that $c = a - 1$.

Therefore, $\mathbf{v}$ can be any vector of the form $\mathbf{v} = \langle a, 2a - 5, a - 1 \rangle$.

b) Explain why there is no vector $\mathbf{v}$ such that $\langle 1, 2, 1 \rangle \times \mathbf{v} = \langle 3, 1, 5 \rangle$.

$$\text{Suppose that } \mathbf{v} = \langle a, b, c \rangle, \text{ then } \langle 1, 2, 1 \rangle \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 1 \\ a & b & c \end{vmatrix} = \langle 2c - b, -(c - a), b - 2a \rangle.$$

$$\text{In order for } \langle 2c - b, -(c - a), b - 2a \rangle = \langle 3, 1, 5 \rangle \text{ we must have } \begin{cases} 2c - b = 3 \\ -(c - a) = 1 \\ b - 2a = 5 \end{cases}$$

$$\text{By adding the first and third equations we get } \begin{array}{r} 2c - b = 3 \\ + b - 2a = 5 \\ \hline 2c - 2a = 8 \end{array} \rightarrow c - a = 4.$$

$$\text{By adding this new equation to the second equation we get } \begin{array}{r} c - a = 4 \\ + -(c - a) = 1 \\ \hline 0 = 5 \end{array}.$$

This implies that the system has no solutions.

9. Let P be a point not on the line L that passes through the points Q and R . Show that the distance from point P to the line L is $d = \frac{\|\mathbf{a} \times \mathbf{b}\|}{\|\mathbf{a}\|}$, where $\mathbf{a} = \overrightarrow{QR}$ and $\mathbf{b} = \overrightarrow{QP}$.

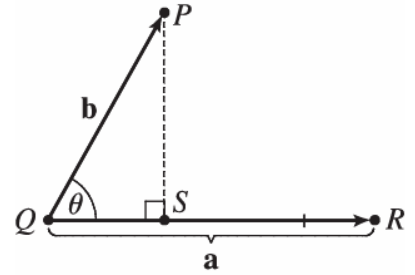
We create a sketch of the situation like shown that the right.

The distance that we are after is $d = \|\overrightarrow{PS}\|$.

We see, by right triangle trig, that $d = \|\overrightarrow{PS}\| = \|\mathbf{b}\| \sin(\theta)$.

We know that $\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin(\theta) \rightarrow \frac{\|\mathbf{a} \times \mathbf{b}\|}{\|\mathbf{a}\| \|\mathbf{b}\|} = \sin(\theta)$.

Therefore, $d = \|\overrightarrow{PS}\| = \|\mathbf{b}\| \left(\frac{\|\mathbf{a} \times \mathbf{b}\|}{\|\mathbf{a}\| \|\mathbf{b}\|} \right) = \frac{\|\mathbf{a} \times \mathbf{b}\|}{\|\mathbf{a}\|}$



10. If $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$, show that $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a}$. Use properties of the cross product to help show that $\mathbf{a} \times \mathbf{b}$ is equal to both $\mathbf{b} \times \mathbf{c}$ and then that $\mathbf{b} \times \mathbf{c}$ is equal to $\mathbf{c} \times \mathbf{a}$.

Because $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$, we know that $\mathbf{b} = -(\mathbf{a} + \mathbf{c})$ we have that

$$\mathbf{a} \times \mathbf{b} = \mathbf{a} \times (-(\mathbf{a} + \mathbf{c})) = -\mathbf{a} \times (\mathbf{a} + \mathbf{c}) = -[\mathbf{a} \times \mathbf{a} + \mathbf{a} \times \mathbf{c}] = -[\mathbf{0} + (\mathbf{a} \times \mathbf{c})] = -(\mathbf{a} \times \mathbf{c}) = \mathbf{c} \times \mathbf{a}$$

Similarly, because $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$, we know that $\mathbf{a} = -(\mathbf{b} + \mathbf{c})$ we have that

$$\mathbf{a} \times \mathbf{b} = -(\mathbf{b} + \mathbf{c}) \times \mathbf{b} = -[\mathbf{b} \times \mathbf{b} + \mathbf{c} \times \mathbf{b}] = -[\mathbf{0} + (\mathbf{c} \times \mathbf{b})] = -(\mathbf{c} \times \mathbf{b}) = \mathbf{b} \times \mathbf{c}$$

Therefore, we have that $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a}$.

11. Prove that $(\mathbf{a} - \mathbf{b}) \times (\mathbf{a} + \mathbf{b}) = 2(\mathbf{a} \times \mathbf{b})$.

$$\begin{aligned} (\mathbf{a} - \mathbf{b}) \times (\mathbf{a} + \mathbf{b}) &= (\mathbf{a} - \mathbf{b}) \times \mathbf{a} + (\mathbf{a} - \mathbf{b}) \times \mathbf{b} \\ &= (\mathbf{a} \times \mathbf{a}) - (\mathbf{b} \times \mathbf{a}) + (\mathbf{a} \times \mathbf{b}) - (\mathbf{b} \times \mathbf{b}) \\ &= \mathbf{0} + (\mathbf{a} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{b}) - \mathbf{0} \\ &= 2(\mathbf{a} \times \mathbf{b}) \end{aligned}$$

12. Suppose that $\mathbf{a} \neq \mathbf{0}$.

a) If $\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{c}$, does it follow that $\mathbf{b} = \mathbf{c}$?

It does not follow that $\mathbf{b} = \mathbf{c}$. If $\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{c}$ then $\mathbf{a} \cdot \mathbf{b} - \mathbf{a} \cdot \mathbf{c} = 0 \rightarrow \mathbf{a} \cdot (\mathbf{b} - \mathbf{c}) = 0$ which implies that $\mathbf{a}$ and $\mathbf{b} - \mathbf{c}$ are orthogonal. For example, if $\mathbf{a} = \langle 1, 1 \rangle$, $\mathbf{b} = \langle 2, -2 \rangle$, & $\mathbf{c} = \langle 3, -3 \rangle$, then $\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{c}$ but $\mathbf{b} \neq \mathbf{c}$.

b) If $\mathbf{a} \times \mathbf{b} = \mathbf{a} \times \mathbf{c}$, does it follow that $\mathbf{b} = \mathbf{c}$?

It does not follow that $\mathbf{b} = \mathbf{c}$. If $\mathbf{a} \times \mathbf{b} = \mathbf{a} \times \mathbf{c}$ then $\mathbf{a} \times \mathbf{b} - \mathbf{a} \times \mathbf{c} = \mathbf{0} \rightarrow \mathbf{a} \times (\mathbf{b} - \mathbf{c}) = \mathbf{0}$ which implies that $\mathbf{a}$ and $\mathbf{b} - \mathbf{c}$ are parallel. For example, if $\mathbf{a} = \langle 1, 0, 0 \rangle$, $\mathbf{b} = \langle 0, 1, 0 \rangle$, & $\mathbf{c} = \langle 2, 1, 0 \rangle$, then $\mathbf{a} \times \mathbf{b} = \mathbf{a} \times \mathbf{c}$ but $\mathbf{b} \neq \mathbf{c}$.

c) If $\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{c}$ and $\mathbf{a} \times \mathbf{b} = \mathbf{a} \times \mathbf{c}$, does it follow that $\mathbf{b} = \mathbf{c}$?

Yes, this does follow. If $\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{c}$ and $\mathbf{a} \times \mathbf{b} = \mathbf{a} \times \mathbf{c}$, then based on what we have shown in the previous parts we know that $\mathbf{a}$ and $\mathbf{b} - \mathbf{c}$ are orthogonal and also that $\mathbf{a}$ and $\mathbf{b} - \mathbf{c}$ are parallel. Since we know that $\mathbf{a} \neq \mathbf{0}$ it must be the case that $\mathbf{b} - \mathbf{c} = \mathbf{0}$ since the zero vector is the only vector that can be both parallel and perpendicular to another vector. Therefore, if $\mathbf{b} - \mathbf{c} = \mathbf{0}$, then $\mathbf{b} = \mathbf{c}$.